SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: M. Phanith Chandu Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I M. Phanith Chandu (*192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Phanith Chandu

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Analyse the Difference Between Training and Testing MSE

Question

A neural network predicts student scores with an **MSE of 0.85 during training** and **5.20 during testing**. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

1. Introduction

A Neural Network is a machine learning model that learns patterns from training data to make predictions on new data. The performance of the model is evaluated using **Mean Squared Error (MSE)**. A lower MSE value indicates better prediction accuracy. Ideally, both the training and testing MSE values should be close to each other. A large difference between them indicates that the model is unable to generalize well.

2. Given

Training Mean Squared Error (MSE) = **0.85**

Testing Mean Squared Error (MSE) = **5.20**

Performance Metric = **Mean Squared Error (MSE)**

3. Objective

To analyse the difference between the training and testing MSE values, identify the possible reasons for the difference, and suggest suitable techniques to improve the performance of the neural network.

4. Theory

The Mean Squared Error (MSE) measures the average squared difference between the actual values and the predicted values.

Formula

[
MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y_i})^2
]

where,

- **n** = Number of samples
- **yi** = Actual value
- **ŷi** = Predicted value

A smaller MSE indicates better prediction performance.

5. Python Code

```
training_mse = 0.85
testing_mse = 5.20

print("Training MSE :", training_mse)
print("Testing MSE  :", testing_mse)

if testing_mse > training_mse:
    print("\nModel Analysis")
    print("The model is overfitting.")
    print("Training performance is good, but testing performance is poor.")
else:
    print("The model is performing well.")
```

6. Output

```
Training MSE : 0.85
Testing MSE  : 5.20

Model Analysis
The model is overfitting.
Training performance is good, but testing performance is poor.
```

7. Analysis

The training MSE (**0.85**) is much lower than the testing MSE (**5.20**). This indicates that the neural network performs well on the training dataset but fails to predict accurately on unseen data. Therefore, the model is suffering from **overfitting**.

Possible Reasons

- The model has too many hidden layers or neurons.

- The training dataset is small.
- The model has been trained for too many epochs.
- The dataset contains noisy or irrelevant features.
- Regularization techniques have not been applied.

8. Corrective Measures

- Increase the amount of training data.
- Apply **Dropout** layers.
- Use **L1 or L2 Regularization**.
- Implement **Early Stopping**.
- Reduce the complexity of the neural network.
- Perform feature selection and proper data preprocessing.
- Use cross-validation to improve model generalization.

9. Result

The large difference between the training MSE (**0.85**) and testing MSE (**5.20**) shows that the neural network is **overfitting**. The model learns the training data effectively but does not perform well on unseen testing data.

10. Conclusion

The comparison of training and testing MSE values indicates that the neural network has poor generalization capability due to overfitting. By applying regularization techniques, increasing the training data, simplifying the model architecture, and using early stopping, the testing performance can be improved, resulting in a more accurate and reliable predictive model.

This is the format most engineering colleges expect for **10-mark AI/ML assignment questions** because it includes theory, practical implementation, analysis, and conclusion in a well-organized manner.